

Integration bee questions

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Round 1

1. $\int (x - 1)(x + 9)^{10} dx$

2. $\int \sqrt{1 + 3x} dx$

3. $\int \frac{x^2 - 1}{x - 1} dx$

4. $\int \cos^4(x) - \sin^4(x) dx$

5. $\int \frac{1}{x^2+3x+2} dx$

6. $\int \frac{\tan(x)}{\cos^3(x)} dx$

Round 2

1. $\int x^2 \ln(x) \, dx$

2. $\int \frac{x^3}{x^2+1} \, dx$

3. $\int \frac{2(\sin(x)+\cos(x))}{\sin(2x)+\cos(2x)+1} \, dx$

4. $\int \frac{1}{1+\tan(x)} \, dx$

5. $\int \sqrt{1-x^2} \, dx$

6. $\int \frac{dx}{1+e^x}$

7. $\int \frac{\sin^2 x}{\cos^4 x} \, dx$

Finals

1. $\int e^{\sqrt{x}} dx$

2. $\int \frac{x}{9+4x^4} dx$

3. $\int \ln(2+x^2) dx$

Round 1 answers

1. $\frac{(x+9)^{12}}{12} - \frac{10(x+9)^{11}}{11} + c$

2. $\frac{2}{9}(1+3x)^{\frac{3}{2}} + c$

3. $\frac{x^2}{2} + x + c$

4. $\ln(x+1) - \ln(x+2) + c = \ln\left(\frac{x+1}{x+2}\right) + c$

5. $\int (\cos^2(x) - \sin^2(x))(\cos^2(x) + \sin^2(x)) dx$
 $\int (\cos^2(x) - \sin^2(x)) dx$
 $\int \cos(2x) dx$
 $\frac{1}{2} \sin(2x) + c$ or $\sin(x) \cos(x) + c$

6. $\frac{1}{3 \cos^3(x)} + c$

Round 2 answers

1. $\frac{x^3}{3} \ln(x) - \frac{x^3}{9} + c$

2. Use Camel Principle additively: $\int \frac{x^3+x}{x^2+1} - \frac{x}{x^2+1} dx$

$$\int \frac{x(x^2+1)}{x^2+1} dx - \frac{1}{2} \int \frac{2x}{x^2+1} dx$$

$$\int x dx - \frac{1}{2} \int \frac{2x}{x^2+1} dx$$

$$\frac{x^2}{2} - \frac{1}{2} \ln(x^2 + 1) + c$$

3. $\ln |\sec(x) + \tan(x)| + c$

4. $\int \frac{1}{1+\frac{\sin x}{\cos x}} dx$

$$\int \frac{1}{\frac{\cos x + \sin x}{\cos x}} dx$$

$$\int \frac{\cos x}{\sin x + \cos x} dx$$

$$\frac{1}{2} \int \frac{2 \cos x}{\sin x + \cos x} dx$$

$$\frac{1}{2} \int \frac{2 \cos x + \sin x - \sin x}{\sin x + \cos x} dx$$

$$\frac{1}{2} \int \left(\frac{\cos x + \sin x}{\sin x + \cos x} + \frac{\cos x - \sin x}{\sin x + \cos x} \right) dx$$

$$\frac{1}{2} \int 1 dx + \frac{1}{2} \int \frac{\cos x - \sin x}{\sin x + \cos x} dx$$

$$\frac{x}{2} + \frac{1}{2} \ln |\sin x + \cos x| + c$$

5. $\frac{x\sqrt{1-x^2}}{2} + \frac{\sin^{-1}(x)}{2} + c$

6. Use Camel Principle additively:

$$\int \frac{1+e^x}{1+e^x} dx - \int \frac{e^x}{1+e^x} dx$$

$$\int 1 dx - \int \frac{e^x}{1+e^x} dx$$

$$x - \ln(1 + e^x) + c$$

Or use Camel Principle multiplicatively:

$$\int \frac{e^{-x}}{e^{-x}+1} dx$$

$$- \int \frac{-e^{-x}}{e^{-x}+1} dx$$

$$- \ln(e^{-x} + 1) + c$$

$$7. \int \frac{\sin^2 x}{\cos^4 x} dx = \int \tan^2 x \sec^2 x dx$$

$$u = \tan x$$

$$du = \sec^2 x dx$$

$$\int u^2 du = \frac{u^3}{3} + c$$

$$= \frac{\tan^3 x}{3} + c$$

Finals answers

1. $u = \sqrt{x}$

$$u^2 = x$$

$$2u \, du = dx$$

$$2 \int u e^u \, du$$

Using Integration by Parts gives:

$$2e^{\sqrt{x}}(\sqrt{x} - 1) + c$$

2. Using a trig substitution we rewrite the integral as:

$$\frac{3}{4} \int \frac{\sec^2(\theta)}{9+9 \tan^2(\theta)} \, d\theta$$

$$\frac{3}{4} \int \frac{\sec^2(\theta)}{9(1+\tan^2(\theta))} \, d\theta$$

$$\frac{3}{4} \int \frac{\sec^2(\theta)}{9(\sec^2(\theta))} \, d\theta$$

$$\frac{1}{12} \int 1 \, d\theta$$

$$\frac{1}{12} \theta + c$$

$$\frac{1}{12} \tan^{-1}\left(\frac{2x^2}{3}\right) + c$$

3. Integrate by parts, setting $f' = 1$ and $g = \ln(2 + x^2)$

$$\int \ln(2 + x^2) \, dx = x \ln(2 + x^2) - \int \frac{2x^2}{2+x^2}$$

Integrate the second part by using a trig substitution:

$$\tan(\theta) = \frac{x}{\sqrt{2}}$$

After simplification we get:

$$2\sqrt{2} \int \tan^2(\theta) \, d\theta$$

$$2\sqrt{2} \int \sec^2(\theta) - 1 \, d\theta$$

$$2\sqrt{2}\tan(\theta) - 2\sqrt{2}\theta + c$$

Reversing substitution:

$$2x - 2\sqrt{2}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + c$$

Combining with first part of original integral, the answer is:

$$x\ln(2+x^2) - 2x + 2\sqrt{2}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + c$$